Video-Conditioned Flow Matching for Human Trajectory Prediction: Progress Report (Bilan)

Dhiaa Eddine Bahri

October 11, 2026

Abstract

This report summarizes the work carried out so far on extending *MoFlow*, a conditional flow-matching framework for multi-modal human trajectory prediction, with dual-stream video conditioning. Three Teacher models are compared on the Stanford Drone Dataset (SDD): a kinematics-only baseline, a model conditioned on global scene video features, and a model further conditioned on agent-centric visual crops. Preliminary single-configuration results indicate that the global video encoder consistently reduces average displacement error (-11.8% on ADE_{avg}) and predictive variance, whereas the agent-centric branch degrades mean errors but improves best-of-K and joint final-metric performance. Next steps are to (i) correlate these patterns with quantitative scene-complexity statistics computed by a dedicated standalone script, and (ii) distill the strongest video-conditioned Teacher into a one-step IMLE Student to retain the speed benefit of MoFlow.

1 Baseline Model Architecture

The starting point of this work is *MoFlow*, a generative trajectory forecaster built on conditional flow matching (CFM). The core idea is to approximate a target vector field that transports samples from a Gaussian prior to the empirical distribution of future motions, conditioned only on past kinematics and on the social context between agents.

Flow-matching formulation. Let $Y^0 \sim \mathcal{N}(0, I)$ be a noise variable and Y^1 the observed future trajectory. A linear probability path is defined by

$$Y^t = (1-t)\,Y^0 + t\,Y^1, \qquad U^t = Y^1 - Y^0, \qquad t \in [0, 1), \tag{1}$$

and a network v_θ is trained to regress the target velocity field U^t. In practice the regression is re-parameterized in data space (as in the original paper) through $D_\theta(Y^t, C, t) = Y^t + (1-t)\,v_\theta(Y^t, C, t)$, leading to the equivalent objective

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{Y^t, Y^1, t}\left[\frac{\left\|D_\theta(Y^t, C, t) - Y^1\right\|_2^2}{(1-t)^2}\right], \tag{2}$$

where C denotes the contextual input and t the flow time. To favour multi-modal and *correlated* predictions, the decoder emits K scene-level waypoint proposals $\{S_i\}_{i=1}^K$, $S_i \in \mathbb{R}^{A \times 2T_f}$, together with classification logits $\{\zeta_i\}_{i=1}^K$; training applies a "closest-prediction" (winner-takes-all) objective

$$\bar{\mathcal{L}}_{\text{FM}} = \mathbb{E}\Big[\left\|S_{j^*} - Y^1\right\|_2^2 + \text{CE}(\zeta_{1:K}, j^*)\Big], \qquad j^* = \arg\min_j \left\|S_j - Y^1\right\|_2^2. \tag{3}$$

A logit-normal schedule is used for the flow time ($\mu_t = -0.5$, $\sigma_t = 1.5$) instead of a uniform one, as it concentrates training in the noisier regime $t \approx 0$ where the velocity field is harder to estimate. Inference solves the resulting ODE with Euler steps.

Context encoder. Historical states $C \in \mathbb{R}^{A \times T_{\text{obs}} \times d}$ are processed by a per-agent MLP into initial features $H_0 \in \mathbb{R}^{A \times d}$, augmented with agent-specific positional encodings $H_0' = H_0 + P_{EA}$, and refined by multi-head self-attention across agents:

$$H_{\text{enc}} = \text{MHSA}\big(Q = H_0',\ K = H_0',\ V = H_0'\big), \tag{4}$$

so that each agent's representation integrates the motion of its neighbours. The output $H_{\text{enc}} \in \mathbb{R}^{A \times d}$ is passed to the motion decoder that performs the flow regression above. This baseline is *kinematics-only*: all visual information in the scene is ignored, and the future distribution is inferred solely from positions and interactions.

Evaluation protocol (SDD). Following the original paper, the Stanford Drone Dataset is used with 8 observed frames (3.2 s) and 12 predicted frames (4.8 s); coordinates are bbox-centre pixel positions, so metric values are reported in pixels (with the caveat that the SDD planar assumption makes pixels an approximately metric unit in this setting). Our experimental harness reports the mean/minimum average and final displacement errors, their variance across the K samples, joint analogues, and a dispersion indicator (MASD).

(a) No Video Encoder

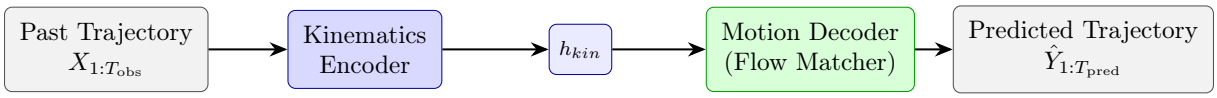


Figure 1: Kinematics-only baseline of the MoFlow Teacher (NV): the past trajectory is compressed by a kinematics encoder into h_{kin}, and the conditional flow-matching decoder regresses the future distribution.

2 Global Video Encoder Architecture

The first extension augments the baseline with *global scene-level visual information*. The rationale is that the past trajectory alone is often ambiguous: the same kinematics can correspond to different futures depending on the surrounding structure (walkable areas, obstacles, roads), and the scene constrains the set of plausible outcomes.

Visual feature extraction. An observed clip $V \in \mathbb{R}^{T_{\text{obs}} \times H \times W \times 3}$ is processed by a 2D ResNet-18 backbone applied frame-wise over the T_{obs} observed frames. This deliberately avoids 3D spatio-temporal convolutions and keeps the temporal alignment with the trajectory stream explicit. The backbone produces per-frame descriptors $z_{\text{vid}} \in \mathbb{R}^{T_{\text{obs}} \times D_{\text{raw}}}$ (typically $D_{\text{raw}} = 512$), which are projected into the model dimension D used throughout the architecture. The backbone is kept small and, in the current implementation, its features are pre-extracted offline and cached so that training remains cheap.

Fusion with the trajectory stream. The video tokens are combined with the trajectory–social tokens z_{traj} through cross-modal attention, in which the trajectory tokens act as queries over the visual tokens:

$$z_{\text{fused}} = \text{CrossAttention}\big(Q = z_{\text{traj}},\ K = z_{\text{vid}},\ V = z_{\text{vid}}\big), \tag{5}$$

and the resulting context $z_{\text{ctx}} = \text{Fusion}(z_{\text{traj}}, z_{\text{fused}})$ is injected into the conditional flow-matching decoder, so that the generated velocity field is explicitly modulated by visual scene structure: $v_\theta\big(Y^t, t \mid z_{\text{traj}}, z_{\text{vid}}\big)$. At the scene level no cross-agent masking is required, since the global representation is shared by all agents.

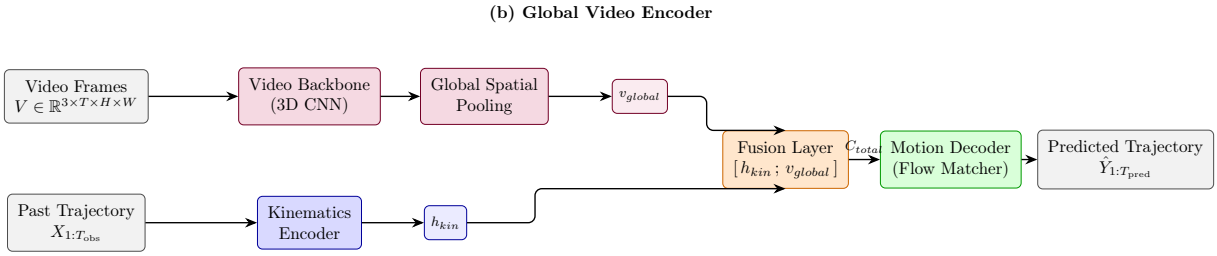


Figure 2: Global-video extension (GV): the observed clip is embedded by the video backbone, pooled into a scene-level token v_{global}, and fused with the trajectory stream before the flow-matching decoder.

3 Results: Baseline vs. Global Video Encoder

Tables 1 present the per-scene comparison between the kinematics-only baseline and the global-video-conditioned model on the eight SDD scenes. Values are scene means over the four reported horizons (labels given in the metrics glossary in Section 6).

Table 1: Per-scene comparison, Baseline ("NV") vs. Global Video Encoder ("GV"). Scene mean over the reported horizons. Lower is better.

Scene	ADE_{avg}		FDE_{avg}		ADE_{min}		FDE_{min}	
	NV	GV	NV	GV	NV	GV	NV	GV
bookstore	12.960	7.981	24.058	14.843	0.675	0.529	1.071	0.744
coupa	6.266	6.421	11.462	11.854	0.386	0.388	0.451	0.446
deathCircle	7.033	7.277	12.760	13.354	0.897	0.995	1.346	1.572
gates	7.642	7.964	14.126	14.675	0.849	0.828	1.149	1.106
hyang	8.629	7.345	15.817	13.513	0.512	0.491	0.666	0.626
little	6.891	6.745	12.770	12.413	0.472	0.471	0.643	0.639
nexus	7.769	6.309	14.502	11.667	0.526	0.513	0.720	0.690
quad	7.545	7.214	13.913	13.290	0.777	0.677	0.949	0.629

Summary. On average-displacement metrics the global video encoder is strictly better than the kinematics-only baseline in **5 of 8** scenes (bookstore, hyang, little, nexus, quad), with the largest gain on *bookstore* (ADE_{avg}: $12.96 \to 7.98$, -38%). The remaining three scenes (coupa, deathCircle, gates) show only a marginal degradation ($\lesssim +3\%$). The minimum-error metrics are improved in 6 of 8 scenes for FDE_{min} (bookstore, coupa, gates, hyang, nexus, quad).

Looking at the scene means over all horizons and scenes, the global branch reduces ADE_{avg} from $5.242 \to 4.624$ (-11.8%) and FDE_{avg} from $9.346 \to 8.266$ (-11.6%), while the predictive variance drops by roughly 16% in both AVar and FVar (Table 2). Qualitatively, the global-conditioned model tends to produce futures that respect the walkable structure of the scene (Figure 3).

4 Agent-Centric Video Encoder Architecture

The second extension addresses the local, per-agent visual cues that a global representation cannot capture: body orientation, posture, immediate neighbours, and the local obstacles right around the pedestrian. These are precisely the cues that help to disambiguate *intention*.

Per-agent crop extraction. For each agent i and each observed frame, the agent's last observed position is projected into pixel space using the scene's homography (world-to-pixel transform retained by the data loader). A square patch $P_i^{(t)} \in \mathbb{R}^{h\times w\times 3}$ (default 64×64 pixels) is cropped around the projected location, with zero-padding at image boundaries. Because crops

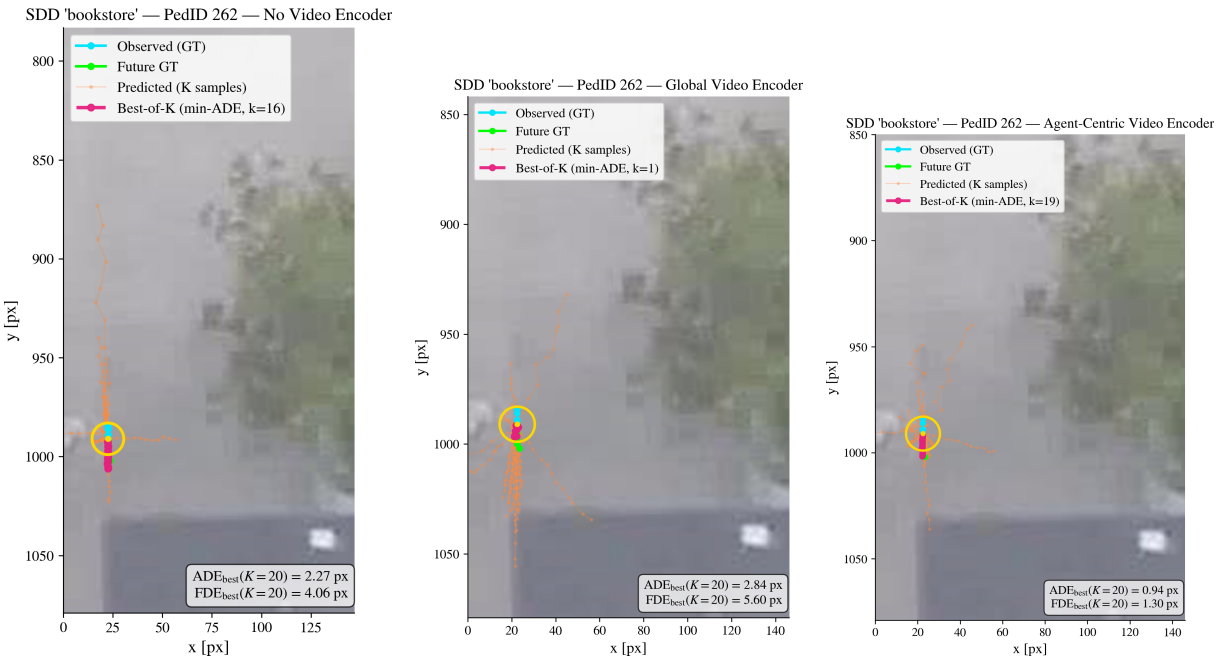


Figure 3: Qualitative comparison on a bookstore window: no-video, global and agent-centric conditioning. In a dense, structured scene the visual branches anchor the sampled futures to the walkable area.

are computed from the same coordinates that produce the trajectory entries, the agent-ID and temporal-index arrays remain perfectly aligned with the crop stack — a strict requirement of the implementation.

Encoding. The per-agent clip $V_i \in \mathbb{R}^{T_{\text{obs}} \times 3 \times h \times w}$ is encoded by a shared 2D ResNet-18 (the first two blocks are kept frozen, with spatial dropout applied after the backbone) into agent-specific tokens $z_{\text{ag}}^{(i)} \in \mathbb{R}^{T_{\text{obs}} \times D}$. Thus the output stream has shape $[B, A, T_{\text{obs}}, D]$ and is subsequently pooled into one token per agent.

Tri-modal fusion. The agent-centric tokens are combined with the two other modalities through a cascade of two cross-attention steps, in the spirit of "flash"-style agents-to-local-then-global attention:

1. *Local step (A):* each agent's trajectory token queries its own agent-video feature, with a strict *identity mask* so that no information leaks between different agents;

2. *Global step (B):* the locally fused features additionally query the global scene tokens from Section 2.

Both attention blocks operate in the canonical hidden dimension D, keeping the tensor-shape contract of the architecture unchanged. The final context is still $[B, A, D]$ and feeds the same motion decoder.

5 Results: Including Agent-Centric Video

Adding the agent-centric branch (and keeping the global branch) produces a markedly different, and more contrasted, pattern.

The global-video advantage on the *average* metrics is lost: the agent-centric model is worse than the baseline in 7 of 8 scenes on ADE_{avg} (Table 2; e.g. *little* 6.891 → 11.001, *quad* 7.545 → 9.986), and its variance indicators AVar/FVar increase substantially (Table 2).

Strikingly, however, the *best-of-K* family paints the opposite picture (Table 3): the agent-centric model wins on FDE_{min} in 3 of 8 scenes, on JADE_{min} in 4 of 8, and on JFDE_{min} in

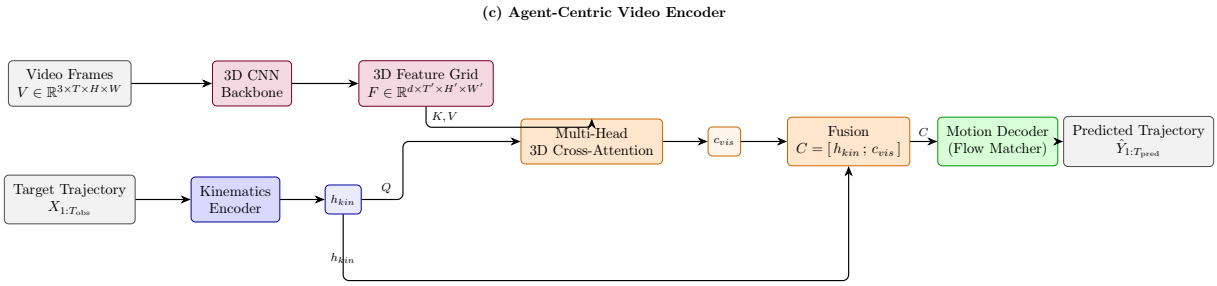


Figure 4: Agent-centric extension (AV): per-agent crops around the target are encoded into local tokens that condition a cross-attention where the kinematic state acts as query; the locally fused context c_{vis} is then combined with the trajectory stream before the flow-matching decoder.

Table 2: Overall means over the eight scenes (scene mean over the reported horizons) for the three Teacher models, and relative change Δ of Global (GV) and Agent-Centric (AV) with respect to No-Video (NV). Lower is better except where it measures dispersion.

Metric	NV	GV	AV	ΔGV	ΔAV
ADE_{avg}	5.242	4.624	5.834	-11.8%	$+11.3\%$
FDE_{avg}	9.346	8.266	10.482	-11.6%	$+12.2\%$
ADE_{min}	0.470	0.457	0.510	-2.8%	$+8.5\%$
FDE_{min}	0.556	0.527	0.576	-5.2%	$+3.6\%$
JADE_{min}	1.229	1.120	1.096	-8.9%	-10.8%
JFDE_{min}	2.058	1.826	1.697	-11.3%	-17.5%
AVar	50.742	42.253	59.451	-16.7%	$+17.2\%$
FVar	133.680	112.607	158.460	-15.8%	$+18.5\%$
MASD	31.295	28.178	32.644	-10.0%	$+4.3\%$

Table 3: Per-scene best-of-K / joint final metrics for the three models. Scene mean over the reported horizons. Lower is better.

	FDE_{min}			JADE_{min}			JFDE_{min}		
Scene	NV	GV	AV	NV	GV	AV	NV	GV	AV
bookstore	1.071	0.744	0.739	2.076	1.141	1.173	3.756	1.999	1.998
coupa	0.451	0.446	0.497	1.200	1.020	0.932	2.171	1.694	1.498
deathCircle	1.346	1.572	1.827	2.261	2.682	2.475	4.000	4.741	4.407
gates	1.149	1.106	1.043	2.577	2.207	2.114	4.393	3.915	3.615
hyang	0.666	0.626	0.822	1.863	1.423	1.468	3.263	2.459	2.508
little	0.643	0.639	0.700	1.472	1.692	1.290	2.626	3.004	2.156
nexus	0.720	0.690	0.489	1.387	1.280	0.992	2.519	2.257	1.663
quad	0.949	0.629	0.847	1.753	1.495	1.698	2.948	2.468	2.626

5 of 8 scenes, with the clearest gains on *nexus* (FDE_{min}: 0.720 → 0.489, −32%; JFDE_{min}: 2.519 → 1.663, −34%), on *coupa* and on *gates*. The textual metrics for the agent-centric model are thus split between a deterioration of the *average* behaviour and an improvement of the *best* behaviour, which is consistent with the accompanying increase in ensemble variance.

Figure 5 summarises the three architectures at a glance.

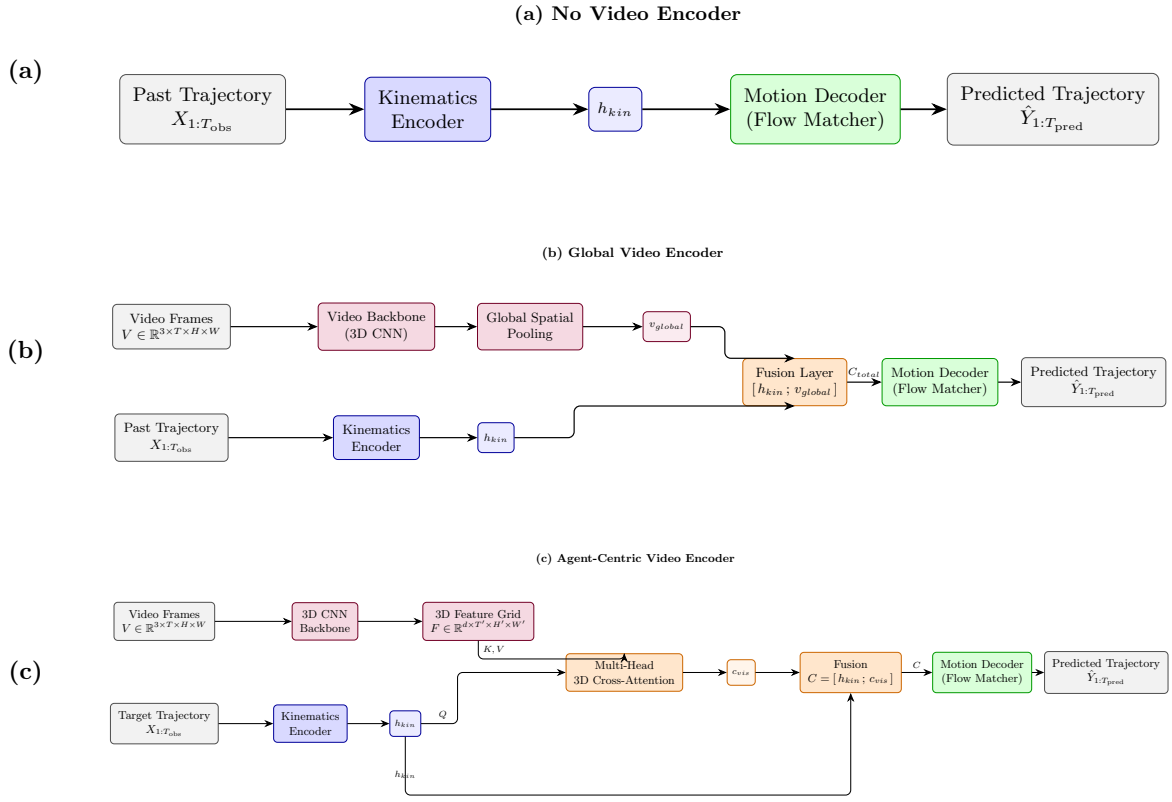


Figure 5: The three Teacher architectures compared: (a) kinematics-only baseline, (b) global-video, and (c) agent-centric conditioning.

6 Interpretation of Results

To interpret the per-scene differences reported above, the intrinsic conditions of each scene were quantified directly from the raw SDD annotations: the crowd density (mean and peak number of simultaneously visible pedestrians), the intensity of pairwise proximity between pedestrians (mean number of close pairs per frame at 25 px and 50 px), the presence and activity of non-pedestrian entities (bikers, skaters, carts, cars, buses), and the mean per-frame displacement of the agents. Table 4 summarises these statistics; they serve in this section as the observed scene facts against which the behaviour of the three models is interpreted.

6.1 Crowd density and social interactions

hyang and *nexus* are by far the most crowded scenes: they average **120.6** and **100.1** simultaneously active pedestrians per frame, peaking at 253 and 256, and exhibit the largest proximity loads, with 31.2 and 22.5 close pedestrian pairs per frame within 25 px and 92.6 and 74.8 pairs within 50 px, respectively. *bookstore* is the densest in per-area terms (0.047 active pedestrians per 10^3 px^2, largest of all scenes) and, with *nexus*, has the highest spatial coverage ($\kappa \approx 0.09$). In

Table 4: Scene-complexity statistics aggregated over all recordings of each SDD scene. $\overline{A}_{\text{ped}}/A_{\max}$: mean/peak active pedestrians per frame; $\text{pairs}_{<d}$: mean number of pedestrian pairs per frame with centres closer than d px; n_{NP}: distinct non-pedestrian tracks; $\overline{A}_{\text{NP}}$: mean active non-pedestrian agents per frame; $\overline{v}$: mean per-frame step length (px/frame); κ: fraction of the scene footprint covered by pedestrian boxes.

Scene	$\overline{A}_{\text{ped}}$	$A_{\max}$	$\text{pairs}_{<25}$	$\text{pairs}_{<50}$	n_{NP}	$\overline{A}_{\text{NP}}$	$\overline{v}$	κ
bookstore	71.5	116	8.4	28.2	363	41.5	0.83	0.092
coupa	40.5	70	2.3	17.5	85	9.9	0.75	0.058
deathCircle	56.6	133	9.5	29.1	1272	113.9	1.26	0.040
gates	48.5	164	5.2	16.0	386	63.7	1.28	0.055
hyang	120.6	253	31.2	92.6	370	52.5	0.84	0.079
little	18.0	58	1.2	4.7	228	25.1	1.39	0.024
nexus	100.1	256	22.5	74.8	270	55.7	0.57	0.094
quad	48.1	51	6.3	23.0	6	8.0	0.92	0.024

these interaction-rich scenes both video branches systematically improve over the kinematics-only baseline, but along *different* metrics.

The global branch produces its largest average gains precisely here: ADE_{avg} drops from 8.63 to 7.35 on *hyang* (-15%), from 7.77 to 6.31 on *nexus* (-19%), and from 12.96 to 7.98 on *bookstore* (-38%), where the predictive variance collapses as well (AVar: 227.8 $\rightarrow$ 107.5, -53%). We attribute this to the global descriptor acting as a *structural prior* that captures the walkable layout of these geometrically constrained scenes: it narrows the generated ensemble around physically admissible regions and reduces the variance of the forecast by roughly 16% overall, exactly the behaviour expected of a conditioning signal that lowers the entropy of the future distribution.

Crowded frames are, however, precisely where a *single pooled* scene embedding is most diluted: in a frame containing hundreds of agents, each individual contributes a negligible fraction of the global descriptor, which therefore encodes "the scene" but nothing about a *specific* agent. This wash-out is visible in the extreme-error statistics: on *nexus* the global model's minimum metrics ($\text{FDE}_{\text{min}} = 0.690$, $\text{JFDE}_{\text{min}} = 2.257$) are superseded by the agent-centric model's 0.489 (FDE_{min}, -32% vs. baseline) and 1.663 (JFDE_{min}, -34%), with $\text{JADE}_{\text{min}} = 0.992$ the best of all variants. The agent-centric branch instead bases its representation on each agent's *own* crop through target-conditioned cross-attention with strict identity masking; it consequently maintains a sharp, focused prediction distribution even at peak crowd density, because the visual evidence attached to the query agent is isolated rather than averaged away. On *hyang* the same pattern holds for the joint final metrics: the agent-centric $\text{JFDE}_{\text{min}} = 2.508$ is within 2% of the global model (2.459) and far below the baseline (3.263, -23%).

The converse is equally instructive. *little*, the least dense scene (18.0 active pedestrians, $0.006\,\text{px}^{-2}$ density, only 1.2 close pairs per frame) and the fastest one ($\overline{v} = 1.39$ px/frame), is the only scene where the agent-centric model degrades the average error markedly (ADE_{avg}: 6.89 $\rightarrow$ 11.00, $+60\%$). When pedestrians are sparse and move quickly, crops frequently miss the target or are dominated by fast occluders, so the local representation injects noise on average — yet the same branch still yields the best JFDE_{min} of any model (2.156, -18% vs. baseline), confirming that its mode sharpness persists whenever the crop is on target.

6.2 The role of heterogeneous and dynamic non-pedestrian obstacles

Every scene contains substantial non-pedestrian traffic, but its composition differs sharply. *deathCircle* is extreme: **1,272** non-pedestrian tracks, among them **914** bikers joined by 106 cars, 128 carts, 106 skaters and 18 buses, with **113.9** such agents active per frame — more than the 56.6 pedestrians — and fast, volatile motion ($\overline{v} = 1.26$, std. 2.23). *nexus* is automotive rather

than two-wheeled: 119 cars and 30 buses among its 270 non-pedestrian tracks, moving slowly (0.57 px/frame, the most congested scene) with up to 256 pedestrians present. *gates* and *little* are biker-dominated (315 and 218 biker tracks), and the latter hosts the fastest pedestrian motion.

Dynamic obstacles cannot be captured by a *static* scene prior: a biker or vehicle traverses the field of view, so any spatially pooled descriptor mixes past and present occupancy and the "walkable area" changes at the frame rate. This is the clearest failure mode of the global branch. On *deathCircle* every model degrades, but the global variant is worst precisely on the min metrics (FDE_{min} 1.572 vs. 1.346 for the baseline; JFDE_{min} 4.741, the largest of all three), consistent with a pooled descriptor saturated by moving clutter. The agent-centric model recovers here by conditioning on the *current* free corridor around each target, which lets the flow matcher tighten its best-of-K modes around feasible pathways and steer them away from collisions (Figure 6). The evidence spans the obstacle-rich scenes: on *nexus* the automotive mix coexists with the largest min-family gains (FDE_{min} -32%, JFDE_{min} -34%); on *gates* with 315 bikers the agent-centric model is the only variant to improve FDE_{min} ($1.149 \rightarrow 1.043$, -9%) and JADE_{min} ($2.577 \rightarrow 2.114$, -18%); on *coupa*, despite its light traffic (85 non-pedestrian tracks), it cuts JADE_{min} by 22% and JFDE_{min} by 31%.

deathCircle also exposes the boundary of the agent-centric design: under such extreme dynamic load, crops are frequently saturated by high-speed distractors, and the average error inflates (FDE_{avg} +16%) as does FDE_{min} (+36%). Yet the branch still keeps the *best* joint-final error among the video variants there (4.407 vs. 4.741 for the global model), i.e. at least one generated mode remains on a feasible trajectory even when the pooled scene descriptor is most diluted. The same mean-vs. mode split is seen on *little* (worst average, best JFDE_{min}), underlining that per-agent visual focus protects the extremes of the distribution even when it cannot protect its mean.

6.3 Trade-off between the global prior and the agent-centric focus

The two video branches are complementary rather than redundant, acting on different parts of the forecast distribution. The **global model acts as a variance-reducing structural prior**: it improves the average metrics in 5 of 8 scenes, wins 5/8 scenes on each of AVar and FVar, and reduces both by $\sim 16\%$ overall. Its agent-agnostic, pooled descriptor regularises the ensemble around walkable structure — lowering the entropy of the generative field — without committing to any single agent; consequently its best-of-K tail is only moderately sharpened (FDE_{min} is best in 4/8 scenes, JADE_{min} in 3/8), because no per-agent evidence exists to anchor a particular mode.

The **agent-centric model is a mode specialist**. By attending only to the target's own crop (identity-masked), it produces sharper commitments: it achieves the best FDE_{min} in 3/8 scenes, JADE_{min} in 4/8 and JFDE_{min} in 5/8, and improves JADE_{min} and JFDE_{min} overall by -10.8% and -17.5% — the largest min-family reductions of any variant. The cost is explicit in the statistics: ensemble variance grows (AVar +17.2%, FVar +18.5%) and the average errors worsen (ADE_{avg} +11.3%). Restricting attention to one agent removes the shared social support that each sample otherwise receives, so individual modes become sharper while the global spread — and with it the mean error — grows.

Taken together, the global branch shapes the *centre and dispersion* of the predicted distribution, whereas the agent-centric branch shapes its *feasible-mode coverage*. The interaction between the two is scene-conditional: in static, geometrically dense scenes such as *bookstore* the global prior dominates; in sparse, fast scenes such as *little* neither branch improves the mean, yet the agent-centric model preserves a good mode; and in congested corridors such as *nexus* and *hyang* the two streams contribute on complementary metrics. This suggests that a merged configuration which keeps the global prior for stability and the agent-centric focus for mode sharpness, with an attention weighting conditioned on scene complexity, is the most promising direction to explore next.

6.4 Status and limitations

These interpretations remain *preliminary*: each model was trained with a single configuration and a single seed on pre-extracted visual features, so the magnitudes of the reported differences must be read as indicative rather than as statistically established. The scene statistics are aggregated over all frames of all recordings of a scene at the raw-annotation rate; speeds are reported in pixels per frame and the pair thresholds (25/50 px) are heuristic. *quad* rests on a very small sample (four recordings, 509 frames, 24 pedestrian and 6 non-pedestrian tracks) and its results should be treated with caution.

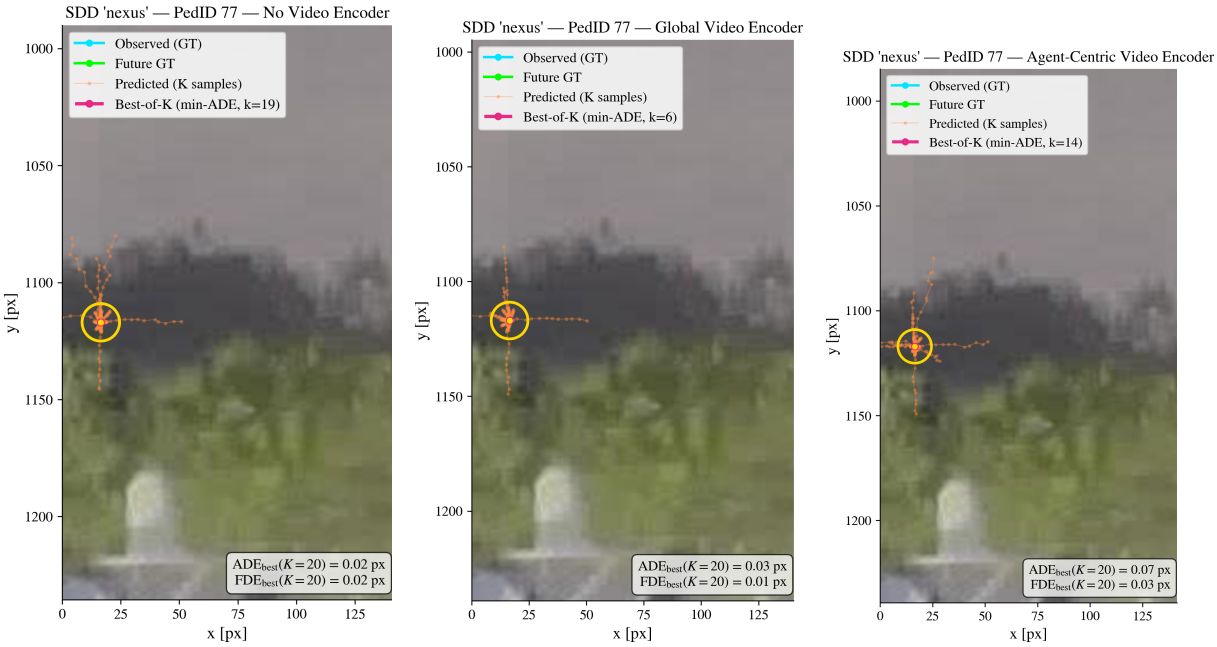


Figure 6: Qualitative comparison on a nexus window: no-video vs. global vs. agent-centric conditioning. In this vehicle-dense scene the agent-centric branch is the one that keeps the predicted futures closest to the observed target.

7 Conclusion and Next Steps

The two video-conditioning extensions of the MoFlow Teacher have been implemented and evaluated on SDD. Global scene conditioning delivers a robust reduction of average error and variance, while agent-centric conditioning shows a complementary and promising improvement of best-of-K behaviour whose source still needs to be understood. The following actions are planned:

1. **Formalise the scene-complexity correlation**: the measured density, interaction and obstacle statistics of Section 6 should be related quantitatively to the per-scene gains (e.g. linear regression of the error reduction against crowd density and non-pedestrian activity) to confirm the qualitative narrative with statistics.

2. **Strengthen the ablations**: train the Global and Agent-Centric variants under a small hyper-parameter sweep (hidden dimension, dropout, crop size, number of frozen backbone blocks) and over several seeds, to assess the stability of the observed effects.

3. **Improve the agent-centric branch**: relax the identity mask to a neighbour-aware local context, add optical-flow/frame-difference inputs for short-term visual motion, and make crop extraction robust to homography error (e.g. multi-scale crops).

4. **Distill the best video-conditioned Teacher into a one-step IMLE Student.** All reported results concern the Teacher; the end-goal remains a video-conditioned Student that preserves the visual gains while restoring MoFlow's 100× faster sampling.

5. **Prepare qualitative case studies** for direction changes and obstacle interactions, comparing no-video vs. video-conditioned futures against the ground truth.